

Hyunseung Hwang

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Education

PhD	KAIST , Electrical Engineering • Advisor: Steven Euijong Whang (Data Intelligence Lab) • Dissertation: Enhancing Explainability in Machine Learning — Explainable Clustering and Explanation Multiplicity	Daejeon, South Korea Sept 2021 – Aug 2026
MS	KAIST , Electrical Engineering	Daejeon, South Korea Sept 2019 – Aug 2021
BS	KAIST , Computer Science	Daejeon, South Korea Sept 2013 – Aug 2019
High School Diploma	Changley High School , General Studies	Fairfax County, VA, USA Sept 2010 – June 2013

Experience

New York University (NYU), Center for Responsible AI , Postdoctoral Researcher • Conducting advanced research on the interpretability and safety of machine learning models under the supervision of Prof. Julia Stoyanovich. • Stress-testing explanation models to ensure safe, confident, and defensible outputs for end-users and stakeholders in high-stakes domains.	Brooklyn, NY, USA June 2026 – present 1 month
Global Frontier Lab , Research Collaborator • Collaborated with Julia Stoyanovich's Responsible AI group at NYU. • Investigated the sensitivity and robustness of AI explainability techniques to deliver safe, confident explanations to stakeholders.	Daejeon, South Korea Sept 2025 – Dec 2025 4 months
Naver , Research Intern • Implemented community detection (Girvan–Newman) on PySpark to cluster unsupervised data for video recommendation.	Seongnam, South Korea June 2019 – Aug 2019 3 months
Kiswe Mobile , Software Engineering Intern • Built Android features for a live video-streaming service using Android Studio. • Updated the application to meet Android Oreo (8.0+) requirements.	Murray Hill, NJ, USA June 2018 – Aug 2018 3 months

Publications

SHAP-based Explanations are Sensitive to Feature Representation	June 2025
Routine data-engineering choices can manipulate SHAP feature importance for the same model — implications for model risk and fair lending. Hyunseung Hwang, Andrew Bell, Joao Fonseca, Venetia Pliatsika, Julia Stoyanovich, Steven Euijong Whang 10.1145/3715275.3732105 (ACM Conference on Fairness, Accountability, and Transparency (FAccT))	

XClusters: Explainability-first Clustering

June 2023

Makes explainability a first-class objective of clustering; evaluated on a real credit-card dataset.

Hyunseung Hwang, Steven Euijong Whang

arxiv.org/abs/2209.10956 (AAAI Conference on Artificial Intelligence (AAAI))

Open-World COVID-19 Data Visualization

Mar 2021

On producing useful visualizations in an open-world setting where the relevant datasets aren't known in advance.

Hyunseung Hwang, Steven Euijong Whang

[10.1007/978-3-030-71055-2_8](https://doi.org/10.1007/978-3-030-71055-2_8) (VLDB Workshop on Data Management and Analytics for Medicine and Healthcare (DMAH))

Skills

Research: Explainable AI (XAI), SHAP / feature attribution, explanation multiplicity and robustness, fairness and responsible AI, explainable clustering

Technical: Python, PySpark, machine learning, data visualization

Languages

Korean: Native speaker

English: Fluent